



Name: _____

Class: _____

James Ruse Agricultural High School

2023 Year 12 Task 3

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 80 minutes
- Write using black pen
- Calculators approved by NESA may be used NESA Reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/ or calculations
- Write your student number at the top of every page of your answer sheets.

Total marks:35

Section I – 5 marks

- Attempt Questions 1–5 on the Multiple Choice Answer Sheet.
- Allow about 7 minutes for this section

Section II – 30 marks

- Attempt Questions 6–10
- **Start EACH Question on a new sheet of paper.**
- Allow about 1 hour and 13 minutes for this section

This paper MUST NOT be removed from the examination room.

Section I

5 marks

Attempt Questions 1 – 5

Allow about 7 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 5.

1. Two forces $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ N and $\begin{pmatrix} -7 \\ -5 \end{pmatrix}$ N act on a particle. What is the resultant force?

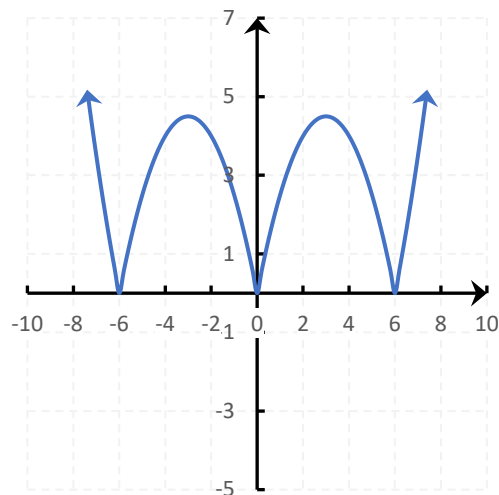
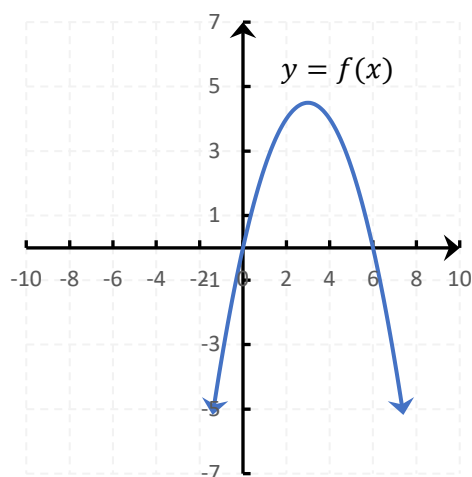
A. $\begin{pmatrix} -21 \\ 10 \end{pmatrix}$ N

B. $\begin{pmatrix} -4 \\ -7 \end{pmatrix}$ N

C. $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ N

D. $\begin{pmatrix} 10 \\ 3 \end{pmatrix}$ N

2. The graph $y = f(x)$ is sketched below, under which transformation will the diagram transform to the one on the right?



- A. $y = |f(x)|$
- B. $y = f(|x|)$
- C. $|y| = f(x)$
- D. $y = |f(|x|)|$

3. Consider a random variable that is normally distributed with a mean of 0 and standard deviation of 1. Define $\Phi(x) = P(X \leq x)$ for $x > 0$. Which of the following is NOT true?

- A. $\Phi(-x) = 1 - \Phi(x)$
- B. $P(|X - x| \leq 1) = \Phi(x + 1) - \Phi(x - 1)$
- C. $P(|X| \leq x) = 2\Phi(x) - 1$
- D. $P(|X| \geq x) = 2 - \Phi(x)$

4. Let $\underline{a}$ and $\underline{b}$ be two-unit vectors and θ is the angle between them. Then $\underline{a} + \underline{b}$ is a unit vector if

- A. $\theta = \frac{\pi}{4}$
- B. $\theta = \frac{\pi}{3}$
- C. $\theta = \frac{\pi}{2}$
- D. $\theta = \frac{2\pi}{3}$

5. Two forces, $F_A = 4\hat{i} - 2\hat{j}$ and $F_B = 2\hat{i} + 4\hat{j}$, act on a particle. The particle is initially at rest at position $\hat{i} + \hat{j}$. All components are measured in newtons and displacements are measured in meters. The Cartesian equation of the path of the particle is

- A. $y = -3x - 1$
- B. $y = \frac{x}{3} - \frac{2}{3}$
- C. $y = 3x + 1$
- D. $y = \frac{x}{3} + \frac{2}{3}$

Section II

42marks

Attempt Questions 6 – 12

Allow about 1 hour and 13 minutes for this section.

Answer each question on a new page. Extra writing pages are available.
For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 **Start on a new page.** **7 marks**

- a) The parametric equations of a curve are given by

$$\begin{aligned} x &= \cos \theta + \sin \theta - 2 \\ y &= \sin 2\theta \end{aligned}, \quad 0 \leq \theta \leq 2\pi$$

- i) Show that the cartesian equation of the curve is 2

$$y = x^2 + 4x + 3$$

- ii) Sketch the curve which corresponds to the above parametric equations, showing the turning points, x- and y- intercepts, and the exact coordinates of the endpoints, if applicable. 2

- b) Consider a random variable that is normally distributed with a mean of 3 and a standard deviation of σ , and $\frac{P(X < 1)}{P(X < 5)} = \frac{1}{9}$. 3
Find the value of $P(|X - 4| < 1)$

Question 7 **Start on a new page.** **7 marks**

- a) Consider the points $A(1, 1)$ and $B\left(\frac{\sqrt{3}}{2} - \frac{1}{2}, \frac{\sqrt{3}}{2} + \frac{1}{2}\right)$.

- i) Find the value of $\angle AOB$ where O is the origin. 2

- ii) Hence, calculate the area of triangle AOB . 2

- b) Let $\underline{a} = \underline{i} + 2\underline{j}$ and $\underline{b} = 2\underline{i} - 4\underline{j}$, where the angle between $\underline{a}$ and $\underline{b}$ is θ . 3
Find the value of $\cos 2\theta$.

Question 8**Start on a new page.****7 marks**

- a) The probability density function of the random variable X is given by **3**

$$f(x) = \begin{cases} \frac{kx}{\sqrt{4-x^2}}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of constant k

- b) The quadrilateral $OABC$ has $\overrightarrow{OA} = 4\mathbf{i} + 2\mathbf{j}$, $\overrightarrow{OB} = 6\mathbf{i} - 3\mathbf{j}$ and $\overrightarrow{OC} = 8\mathbf{i} - 20\mathbf{j}$.
Find $\overrightarrow{AB}$ and hence show that $OABC$ is a trapezium. **2**

- c) The vectors $\mathbf{a} + 3\mathbf{b}$ and $4\mathbf{a} - \mathbf{b}$ are perpendicular, and $|\mathbf{a}| = 2|\mathbf{b}|$. Determine the angle between the nonzero vectors $\mathbf{a}$ and $\mathbf{b}$. **2**

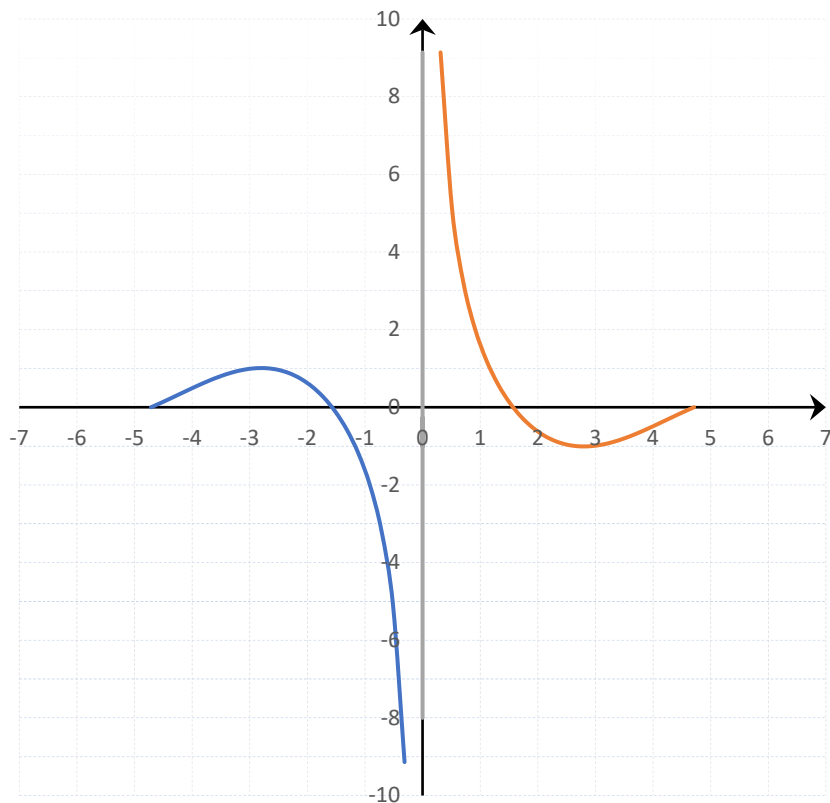
Question 9**Start on a new page.****7 marks**

a) Given the function $y = (x - 1)(|x - 3|)$, $x \in \mathbb{R}$

i) Sketch the function. 2

ii) Hence solve $(x - 1)(|x - 3|) = 1$, leave your answer in exact values. 1

b) The diagram shows the graph of a function $y = f(x)$ over its domain.



Draw the following function on the sheet provided, clearly showing any transformation of the turning points, x - and y - intercepts, and asymptotes, as applicable for:

i) $y = f(|x|)$ 2

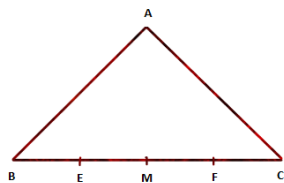
ii) $y = \frac{1}{f(x)}$ 2

Question 10

Start on a new page.

7 marks

- a) In triangle ABC below, given that $|\overrightarrow{BE}| = |\overrightarrow{EM}| = |\overrightarrow{MF}| = |\overrightarrow{FC}|$



Prove that $\overrightarrow{AB} + \overrightarrow{AE} + \overrightarrow{AM} + \overrightarrow{AF} + \overrightarrow{AC} = \frac{5}{2}(\overrightarrow{AB} + \overrightarrow{AC})$ 2

- b) A small airplane leaves an airport on an overcast day and is later sighted 215 km away, in a direction making an angle of $N22^\circ E$ as shown in the diagram. 2

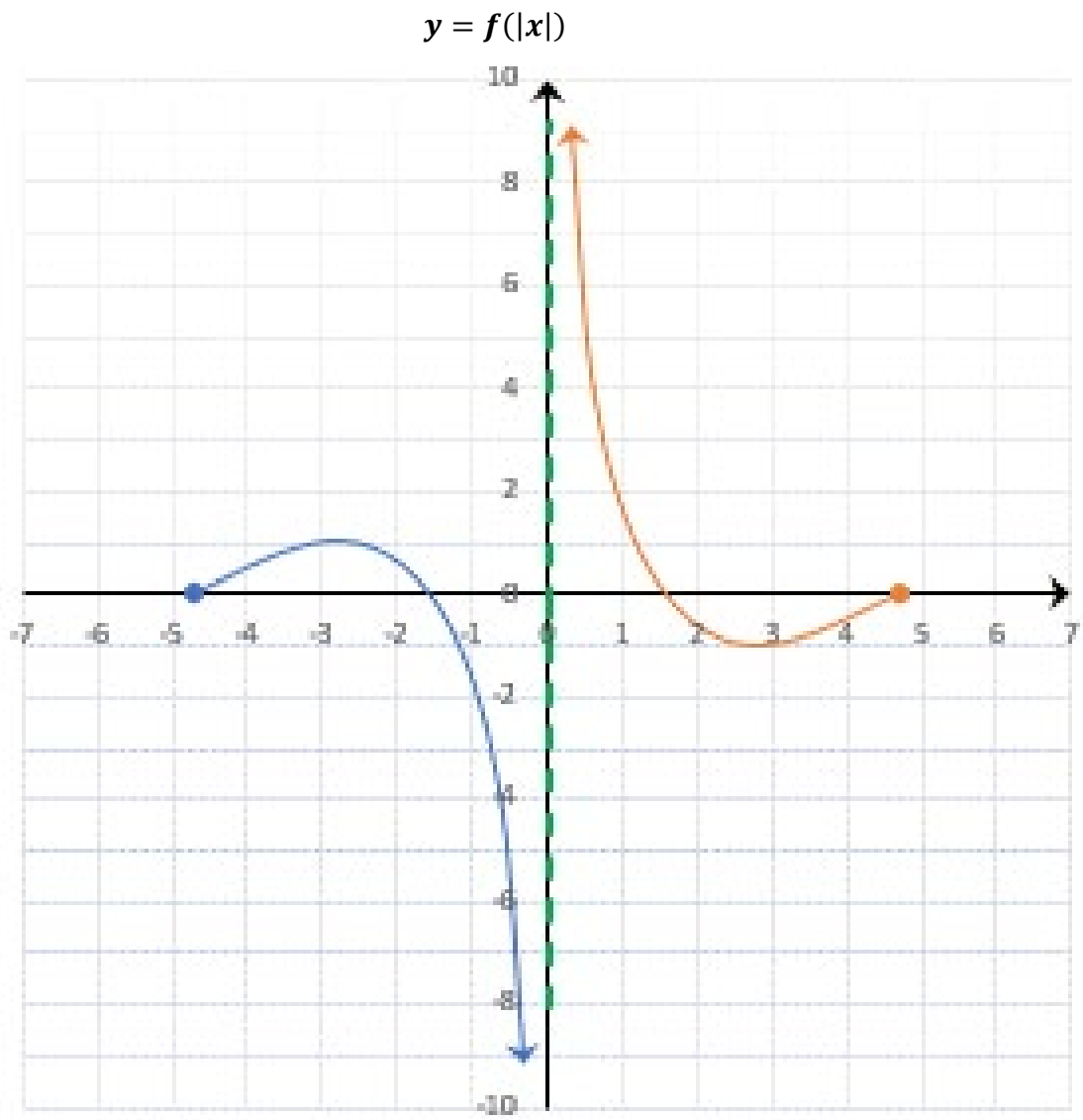
How far east and north is the airplane from the airport when sighted?

- c) The point A and B have respective position vectors $\underline{a}$ and $\underline{b}$, relative to a fixed origin O. The point C lies on $\overrightarrow{AB}$ produced such that $|\overrightarrow{AB}| : |\overrightarrow{AC}| = 1 : 4$. The point D lies on $\overrightarrow{OB}$ produced such that $|\overrightarrow{OB}| : |\overrightarrow{OD}| = 1 : k$, where k is a scalar constant. Given that $\overrightarrow{AB}$ is perpendicular to $\overrightarrow{CD}$, show that: 3

$$k = \frac{3|\underline{a}|^2 - 7\underline{a} \cdot \underline{b} + 4|\underline{b}|^2}{|\underline{b}|^2 - \underline{a} \cdot \underline{b}}$$

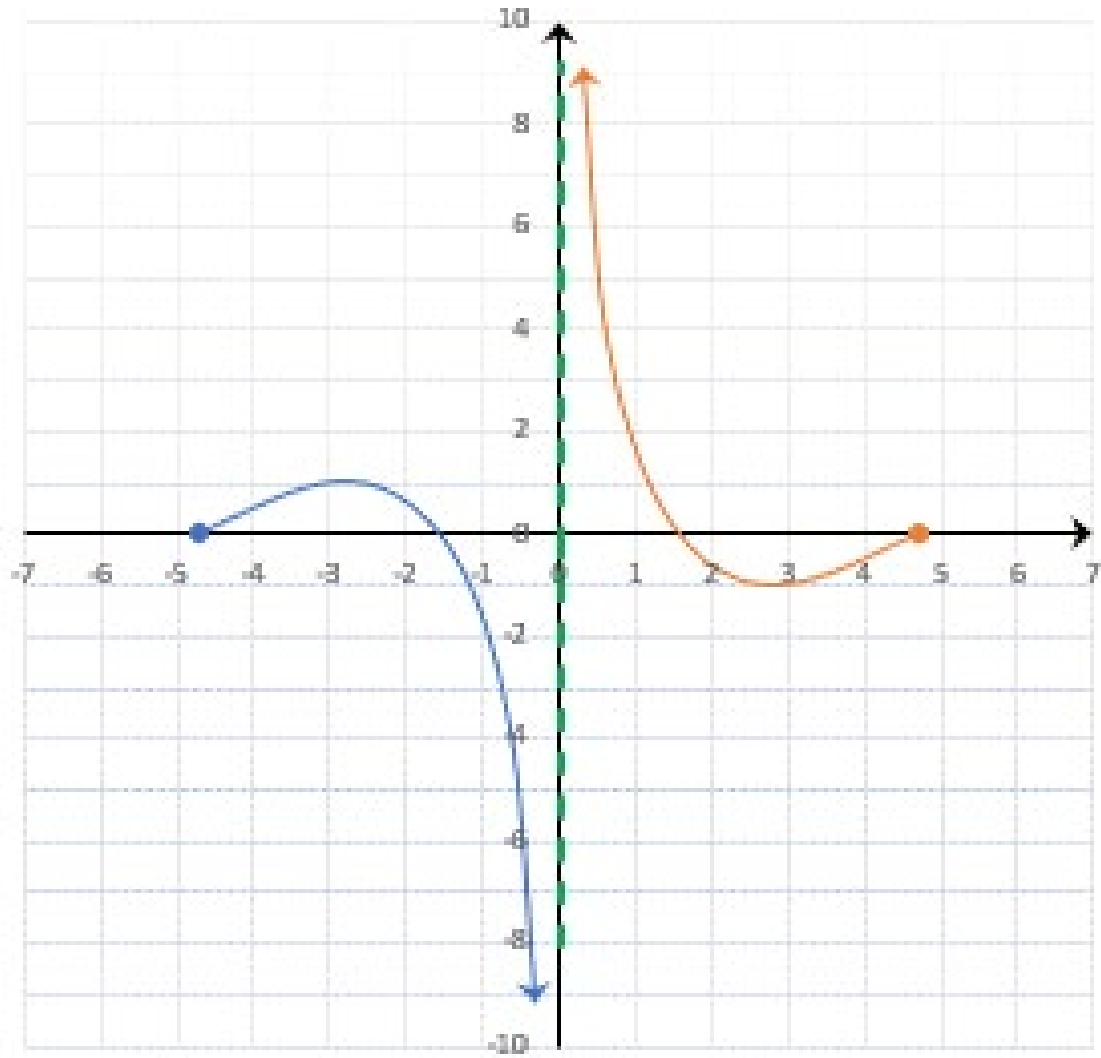
- End of examination -

Question 9b)



Question 9b)

$$y = \frac{1}{f(x)}$$



Section I

Student ID number: |

5 Marks

Attempt Question 1 – 5.

Allow approximately 7 minutes for this section.

Use the multiple choice answer sheet below to record your answers to Question 1 – 5.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct

Year 12 Examination 2023
 Mathematics Extension 1

Multiple Choice Answer Sheet

Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☒	C	☐	D	☐
2	A	☐	B	☐	C	☐	D	☒
3	A	☐	B	☐	C	☐	D	☒
4	A	☐	B	☐	C	☐	D	☒
5	A	☐	B	☐	C	☐	D	☒

Mark: /5

Yr 12 x1 2023 Task 3

(i) $x+2 = \cos \theta + \sin \theta$
 $y = \sin 2\theta = 2 \sin \theta \cos \theta$

$(x+2)^2 = (\cos \theta + \sin \theta)^2$
 1m $(x+2)^2 = \cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta$
 $(x+2)^2 = 1 + y$ ($\cos^2 \theta + \sin^2 \theta = 1$)

$x^2 + 4x + 4 = 1 + y$

1m $\therefore y = x^2 + 4x + 3$ #

Since $y = \sin 2\theta$
 $\therefore -1 \leq y \leq 1$

For $y=1$ $1 = x^2 + 4x + 3$
 $0 = x^2 + 4x + 2$
 $x = \frac{-4 \pm \sqrt{16 - 4(2)}}{2}$

$x = \frac{-4 \pm \sqrt{8}}{2} = \frac{-4 \pm 2\sqrt{2}}{2}$

$x = -2 \pm \sqrt{2}$

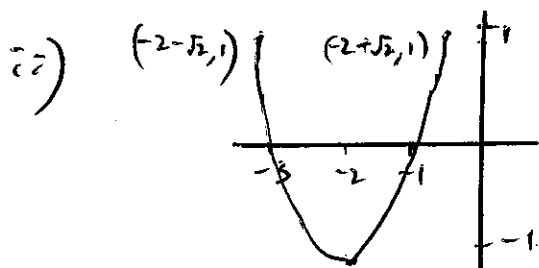
For $y=-1$ $-1 = x^2 + 4x + 3$
 $0 = x^2 + 4x + 4$
 $0 = (x+2)^2$
 $\therefore x = -2$, Vertex at $(-2, -1)$

show Q

must show
 $\sin^2 \theta + \cos^2 \theta = 1$
 $y = 2 \sin \theta \cos \theta$
 $(x+2)^2 = x^2 + 4x + 4$
 otherwise max 1m

1m
 endpts $(-2 \pm \sqrt{2}, 1)$
 or
 parabola with
 some essential points
 TP $(-2, -1)$
 x-intercepts $x = -3, -1$
 endpts $(-2 \pm \sqrt{2}, 1)$

2m perfect graph
 with all essential
 pts



$y = x^2 + 4x + 3$
 $y = (x+3)(x+1)$
 $0 = (x+3)(x+1)$
 $x = -3, -1$ (x-intercepts)

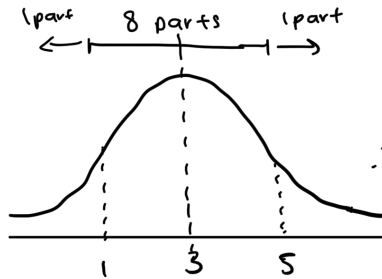
MATHEMATICS Extension 1 : Question..6...

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} \text{b) } P(|x-4| < 1) &= P(-1 < x-4 < 1) \\ &= P(3 < x < 5) \end{aligned}$$



$$= P(x < 5) - P(x < 3)$$

$$\therefore P(x < 5) = \frac{9}{10}$$

OR by symmetry

$$P(x < 1) = 1 - P(x < 5)$$

$$\text{as } 9P(x < 1) = P(x < 5)$$

$$\therefore 9(1 - P(x < 5)) = P(x < 5)$$

$$9 - 9P(x < 5) = P(x < 5)$$

$$9 = 10P(x < 5)$$

$$\therefore P(x < 5) = \frac{9}{10}$$

$$\begin{aligned} \therefore P(|x-4| < 1) &= \frac{9}{10} - \frac{1}{2} \\ &= \frac{2}{5} \end{aligned}$$

1 for getting
 $P(x < 5) = \frac{9}{10}$

Question 7 Ext 1 Task 3, 2023

MATHEMATICS EXTENSION 1 : Question 12		
Suggested Solutions	Marks	Marker's Comments
a) $A(1,1)$ $B(\frac{\sqrt{3}}{2} - \frac{1}{2}, \frac{\sqrt{3}}{2} + \frac{1}{2})$ i) $\angle AOB$? $\vec{OA} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $\vec{OB} = \begin{pmatrix} \frac{\sqrt{3}}{2} - \frac{1}{2} \\ \frac{\sqrt{3}}{2} + \frac{1}{2} \end{pmatrix} = \frac{\sqrt{3}}{2} - \frac{1}{2} + (\frac{\sqrt{3}}{2} + \frac{1}{2})$ $= \sqrt{3}$ $\vec{OA} \cdot \vec{OB} = \vec{OA} \vec{OB} \cos \angle AOB$ $\vec{OA} = \sqrt{1^2 + 1^2} = \sqrt{2}$ $\vec{OB} = \sqrt{(\frac{\sqrt{3}}{2} - \frac{1}{2})^2 + (\frac{\sqrt{3}}{2} + \frac{1}{2})^2} = \sqrt{2}$ $\therefore \angle AOB = \cos^{-1} \frac{\sqrt{3}}{2}$ $= 30^\circ$ ii) Hence calc area of $\triangle AOB$. $A = \frac{1}{2} ab \sin C$ $a = \vec{OA}$, $b = \vec{OB}$ $= \frac{1}{2} (\sqrt{2}) \times (\sqrt{2}) \times \frac{1}{2}$ $= \frac{1}{2} u^2$		1 - angle 1 - $\vec{OA}$ & $\vec{OB}$ & $\vec{OA} \times \vec{OB}$ 1 - answer 1 - correct a & b & formula some students forgot the

Question 7

b)

$$\begin{aligned}\cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 2 \cos^2 \theta - 1\end{aligned}$$

1 mark for correct expansion of $\cos 2\theta$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= (1)(2) + (2)(-4) \\ &= 2 - 8 \\ &= -6\end{aligned}$$

$$\begin{aligned}|\vec{a}| &= \sqrt{1^2 + 2^2} \\ &= \sqrt{5}\end{aligned}$$

$$\begin{aligned}|\vec{b}| &= \sqrt{2^2 + (-4)^2} \\ &= \sqrt{4 + 16} \\ &= \sqrt{20} \\ &= 2\sqrt{5}\end{aligned}$$

1 mark for finding $\vec{a} \cdot \vec{b}$, $|\vec{a}|$, and $|\vec{b}|$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \\ -6 &= \sqrt{5} \times 2\sqrt{5} \times \cos \theta \\ &= 10 \cos \theta \\ \cos \theta &= -\frac{3}{5}\end{aligned}$$

$$\begin{aligned}\therefore \cos 2\theta &= 2 \left(-\frac{3}{5} \right)^2 - 1 \\ &= \frac{18}{25} - 1 \\ &= -\frac{7}{25}\end{aligned}$$

1 mark for final answer

MATHEMATICS EXT 1: Question 8

Suggested Solutions	Marks	Marker's Comments
a, $f(x) = \begin{cases} \frac{kx}{\sqrt{4-x^2}} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$		
Since $f(x)$ is a PDF.		
$\int_0^1 \frac{kx}{\sqrt{4-x^2}} dx = 1$	① Knowing properties of PDF	
$-\frac{1}{2} \int_0^1 -2kx (4-x^2)^{-1/2} dx = 1$		
$-\frac{1}{2} k \left[\frac{(4-x^2)^{1/2}}{1/2} \right]_0^1 = 1$	① Correct integration	
$-k(\sqrt{3} - 2) = 1$		
$k = \frac{1}{2-\sqrt{3}}$ $= 2+\sqrt{3}$	① Final	
COMMENTS:		
dropping the x in the numerator was a common error $\rightarrow$ led to arcsine integral		
Please stop overcomplicating things!! There is no need to choose your own sub or use IBP in an Ext 1 exam $\rightarrow$ if you did this and made an error, no CF marks awarded.		

MATHEMATICS EXT 1: Question

Suggested Solutions	Marks	Marker's Comments
b, $\vec{AB} = \vec{OB} - \vec{OA}$ $= (6i - 3j) - (4i + 2j)$ $= 2i - 5j$ $\vec{OC} = 8i - 20j$ $= 4(2i - 5j)$ $= 4\vec{AB}$ $\therefore \vec{OC} \parallel \vec{AB}$	①	
One pair of sides are parallel, meaning the quadrilateral is a trapezium.	①	
COMMENTS - The majority of students are abusing vector notation. $(2, -5)$ is a point, not a vector. Must have $2i - 5j$ or $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$. - This was marked <u>very</u> leniently $\rightarrow$ defn of trapezium		

Question 8

c)

$$\begin{aligned}
 (\underline{a} \cdot \underline{b}) &= |\underline{a}| |\underline{b}| \cos \theta \\
 &= 2 |\underline{b}|^2 \cos \theta \quad (|\underline{a}| = 2 |\underline{b}|) \\
 (\underline{a} + 3\underline{b}) \cdot (4\underline{a} - \underline{b}) &= 0 \Rightarrow (\underline{a} \cdot 4\underline{a}) - (\underline{a} \cdot \underline{b}) + (3\underline{b} \cdot 4\underline{a}) - (3\underline{b} \cdot \underline{b}) = 0 \\
 4|\underline{a}|^2 - (\underline{a} \cdot \underline{b}) + 12(\underline{a} \cdot \underline{b}) - 3|\underline{b}|^2 &= 0 \\
 4|\underline{a}|^2 + 11(\underline{a} \cdot \underline{b}) - 3|\underline{b}|^2 &= 0 \\
 (\underline{a} \cdot \underline{b}) &= \frac{-4|\underline{a}|^2 + 3|\underline{b}|^2}{11} \\
 \therefore 2|\underline{b}|^2 \cos \theta &= \frac{-4|\underline{a}|^2 + 3|\underline{b}|^2}{11} \\
 \cos \theta &= \frac{-4(2|\underline{b}|)^2 + 3|\underline{b}|^2}{22|\underline{b}|^2} \quad (|\underline{a}| = 2|\underline{b}|) \\
 &= -\frac{13}{22} \\
 \theta &= 126^\circ 13'
 \end{aligned}$$

1 mark for completing **both**:

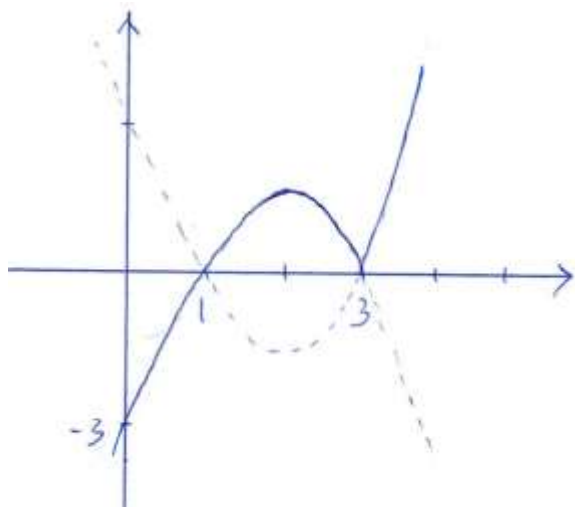
- correctly expanding to $4|\underline{a}|^2 - (\underline{a} \cdot \underline{b}) + 12(\underline{a} \cdot \underline{b}) - 3|\underline{b}|^2 = 0$
- $(\underline{a} \cdot \underline{b}) = |\underline{a}| |\underline{b}| \cos \theta$ or $\cos \theta = \frac{(\underline{a} \cdot \underline{b})}{|\underline{a}| |\underline{b}|}$

1 mark for getting the correct answer. (Obtuse and not acute!)

Students who left their answer as $\theta = \cos^{-1}\left(-\frac{13}{22}\right)$ unfortunately don't receive the 2nd mark.

Year 12 Ext. 1 Task 3 Q9

(a) (i)



1 mark for general shape

1 mark for details – including;

- all three intercepts
- scale on x-axis
- sharp at $x=3$
- correct concavity on both ends

For the responses with bouncing behaviour at $x=3$, 1 mark for the general shape was allowed ONLY IF the symmetry was evident (i.e. clearly shown max at $x=2$)

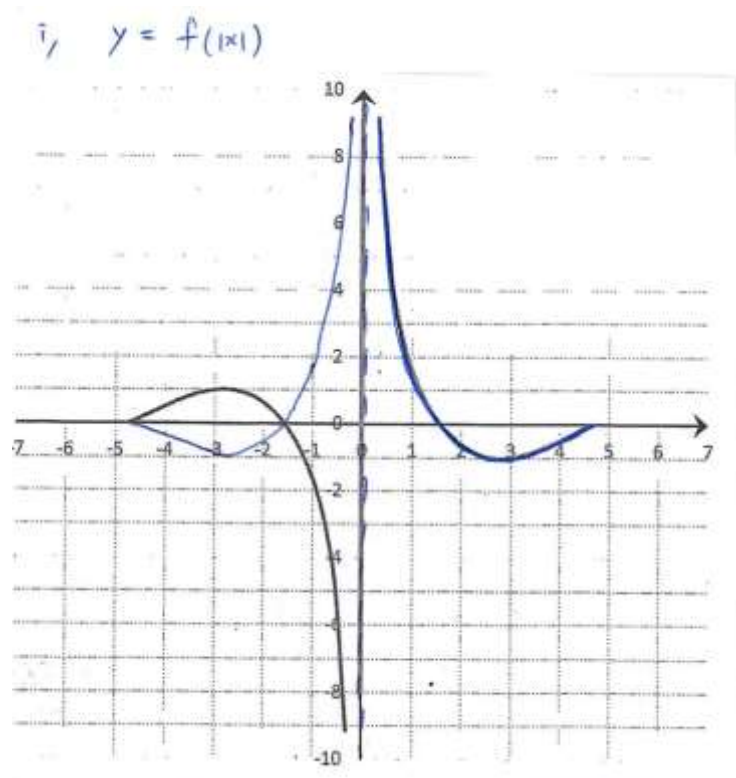
responses with any other wrong graphs awarded zero mark

(a) (ii)

$$x = 2, 2 + \sqrt{2}$$

1 mark for BOTH correct solutions in EXACT values

(b)



1 mark for general shape – must show BOTH branches

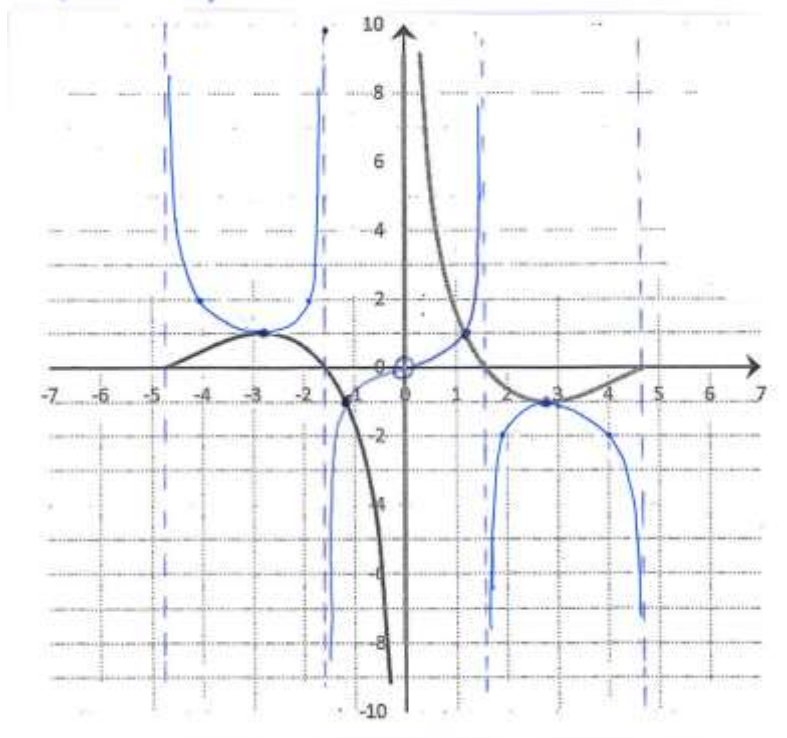
1 mark for details – including intercepts and scales

For the scale, it was particularly considered at;

when $y=2$, the graph must be on the right-hand-side of the grid line (i.e. $x > -1$)

when $y=1$, the graph must be on the left-hand-side of the grid line (i.e. $x < -1$)

ii, $y = \frac{1}{f(x)}$



1 mark for the general shape

1 mark for details – including asymptotes, open-circle at $x=0$ and intersection at $x = \pm 1$

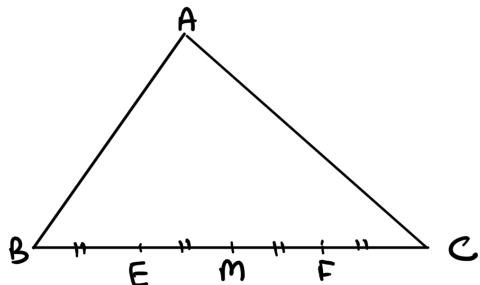
MATHEMATICS Extension 1 : Question 10.

Suggested Solutions

Marks

Marker's Comments

a)



$$\text{RTP: } \vec{AB} + \vec{AE} + \vec{AM} + \vec{AF} + \vec{AC} = \frac{5}{2} (\vec{AB} + \vec{AC})$$

$$\text{LHS} = \vec{AB} + (\vec{AB} + \vec{BE}) + (\vec{AB} + \vec{BM}) + (\vec{AB} + \vec{BF}) + \vec{AC}$$

$$= 4\vec{AB} + \frac{1}{4}\vec{BC} + \frac{1}{2}\vec{BC} + \frac{3}{4}\vec{BC} + \vec{AC}$$

$$\text{as } |\vec{BE}| = |\vec{EM}| = |\vec{MF}| = |\vec{FC}|$$

$$= 4\vec{AB} + \vec{AC} + \frac{3}{2}\vec{BC}$$

$$= 4\vec{AB} + \vec{AC} + \frac{3}{2}(\vec{AC} - \vec{AB})$$

$$= \frac{5}{2}\vec{AB} + \frac{5}{2}\vec{AC}$$

$$= \frac{5}{2}(\vec{AB} + \vec{AC})$$

1 for getting to this step

or any other form.

ie. $5\vec{AB} + 10\vec{BE}$

or $4\vec{AB} + 6\vec{BE} + \vec{AC}$
etc.

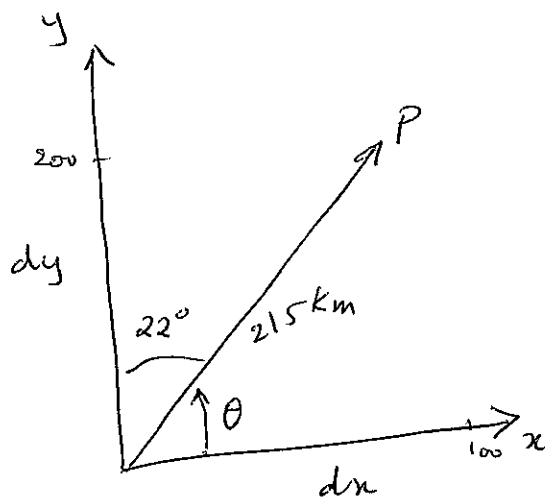
} 1

Question 10

(b) $\theta = 68^\circ$

$$\begin{aligned} dx &= d \cos \theta \\ &= 215 \cos 68 \\ &= 81 \text{ km} \end{aligned} \quad (1 \text{ mark})$$

$$\begin{aligned} dy &= d \sin \theta \\ &= 215 \sin 68 \\ &= 199 \text{ km} \end{aligned} \quad (1 \text{ mark})$$



So the airplane was spotted 199 km North and 81 km East of the airport.

(c) $\underline{a}$, $\underline{b}$ position vectors of A and B

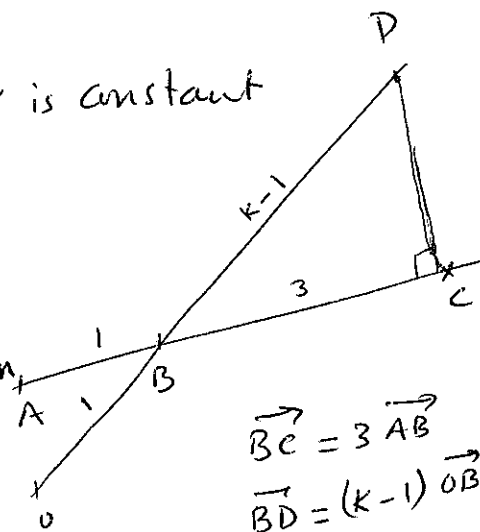
C lies on $\overrightarrow{AB}$, $|\overrightarrow{AB}| : |\overrightarrow{AC}| = 1 : 4$

D lies on $\overrightarrow{OB}$, $|\overrightarrow{OB}| : |\overrightarrow{OD}| = 1 : k$, k is constant

$$\begin{aligned} \overrightarrow{OC} &= 4\overrightarrow{OB} - 3\overrightarrow{OA} \\ &= 4\underline{b} - 3\underline{a} \end{aligned}$$

$$\begin{aligned} \overrightarrow{CD} &= \overrightarrow{BD} - \overrightarrow{BC} \\ &= (k-1)\underline{b} - 3(\underline{b} - \underline{a}) \\ &= (k-4)\underline{b} + 3\underline{a} \end{aligned}$$

(1 mark)
correct expression
for $\overrightarrow{OC}$ or $\overrightarrow{CD}$



$$\begin{aligned} \overrightarrow{BC} &= 3\overrightarrow{AB} \\ \overrightarrow{BD} &= (k-1)\overrightarrow{OB} \end{aligned}$$

$$\overrightarrow{AB} \perp \overrightarrow{CD} \therefore \overrightarrow{AB} \cdot \overrightarrow{CD} = 0$$

$$(\underline{b} - \underline{a}) \cdot ((k-4)\underline{b} + 3\underline{a}) = 0$$

$$(k-4)(\underline{b} \cdot \underline{b}) + 3\underline{a} \cdot \underline{b} - (k-4)\underline{a} \cdot \underline{b} - 3|\underline{a}|^2 = 0$$

$$k(|\underline{b}|^2 - \underline{a} \cdot \underline{b}) = 4|\underline{b}|^2 - 7\underline{a} \cdot \underline{b} + 3|\underline{a}|^2$$

$$k = \frac{4|\underline{b}|^2 - 7\underline{a} \cdot \underline{b} + 3|\underline{a}|^2}{|\underline{b}|^2 - \underline{a} \cdot \underline{b}}$$

(1 mark) progress

(1 mark) work leading to the correct answer